SUPPORTING ADULT GUIDANCE

Zero as a placeholder

Place Value · Lesson 3. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

In the main activity, the Zero Detective, learners choose a number (4,050, 0450, 3.50, 7,006, 20.04 or 100) shown in a place-value chart and tap any zero to take it away. A second chart shows what is left, with any digit that had to slide marked in red, and the lesson says whether the number changed. Six questions follow, and a stretch explains why some zeros can go and others cannot. Learners then write numbers from words, diagnose and fix wrongly written numbers, and build numbers with zeros, including the smallest and largest five-digit numbers from 1, 2 and three zeros.

Driving Question: *When does a zero change a number?*

The mathematics, in plain terms

Zero is a placeholder: it records that a column is empty so that every other digit keeps its place value. A zero is needed exactly when some non-zero digit lies further from the decimal point than it does, on the same side, because that digit would otherwise slide into a new column. Leading zeros of a whole number and trailing zeros after the decimal point change nothing (though in measurement, 3.50 m can signal accuracy; that idea comes later). Decimals are taught properly in Lesson 4.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — A zero that holds a column

What happens if the 0 in the hundreds column of 4,050 is left out?

4,050

Thousands	Hundreds	Tens	Ones
4	0	5	0

Take the hundreds 0 away

Thousands	Hundreds	Tens	Ones
	4	5	0

The 4 slides from the thousands into the hundreds. The number is now 450.

1. 4,050 is	4 thousands, 0 hundreds, 5 tens, 0 ones
2. Without the hundreds 0, the 4 slides one column right.	4 hundreds
3. Read the new number.	450

4,050 and 450 are different numbers. That zero was holding the hundreds column open.

Example 2 — A zero that holds nothing

Is 3.50 the same number as 3.5?

3.50

Ones		Tenths	Hundredths
3	•	5	0

Take the last 0 away

Ones		Tenths	Hundredths
3	•	5	

No other digit moves. The 3 is still 3 ones and the 5 is still 5 tenths.

1. 3.50 is	3 ones, 5 tenths, 0 hundredths
2. Take the last 0 away. Does any digit move?	no
3. So the value is unchanged.	3.5

3.50 = 3.5. The end zero of a decimal has no digits beyond it to hold in place.

Take a zero away and watch, then explain, then use it

Learners see what happens when each zero goes before any rule is stated, so the rule (a zero matters when digits beyond it would slide) is something they observed. Writing numbers from words, diagnosing errors and building numbers then apply the idea in the order it is used.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	The Zero Detective and six questions. Stretch: why can some zeros go?	~16 min
Movement	Words to figures: write each number with its zeros.	~7 min
Reflection	Find and fix: diagnose wrongly written numbers.	~7 min
Creativity	Zero builder: make numbers, including the smallest and largest.	~10 min
Rest	Nothing, holding a place.	~2 min
Nutrition	Intellectual: care with detail; one missing zero changes a number.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Writing “four thousand and fifty” as 450 or 40050.
Ask: “How many columns does four thousand need? Which are empty?”

Writing “three thousand and six” as 3,600.
Ask: “Is the 6 six hundreds or six ones?”

Thinking 0.50 is bigger than 0.5 because it has more digits.
Ask: “How many tenths in each? Does the end zero hold anything?”

Thinking zeros can always be left out because they are worth nothing.
Ask: “Take the zeros out of 6,000. What is left?”

Thinking a front zero (0450) changes the number.
Ask: “Are there any digits to the left of that 0 for it to hold?”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: what each column is worth (Lesson 1) and how digits move (Lesson 2). A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner writes “two million and twenty” as 2,020. The aim is a question that lets them count the columns.

- “How many columns does a million need?”

- “Which column does the 2 of twenty go in?”
- “What goes in the columns in between?”
- “Can you read your number aloud and check it matches?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Should learners say “add a zero”?

No. A zero holds an empty column; the language of holding keeps place value in view.

Are 3.50 and 3.5 always interchangeable?

As numbers, yes. In money (£3.50) and measurement the end zero is a convention or shows accuracy; that is a later idea.

Is the stretch required?

No. The secure core is writing numbers with placeholder zeros correctly and saying whether a given zero is needed.

What comes next?

Lesson 4: beyond the point, tenths, hundredths and thousandths.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.

Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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